# EXPERIMENT REPORT:
# STUDY OF LCR RESONANCE CIRCUITS

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## Abstract

The resonance behavior of a series LCR circuit was investigated by measuring the voltage ratios $V_R/V_i$ and $V_{LC}/V_i$ as functions of frequency. The resonant frequency and Q-factor were obtained experimentally using curve fitting and compared with theoretical values derived from circuit parameters. The experiment confirms the selective response of LCR circuits and illustrates the role of the Q-factor in determining resonance sharpness.

# 1 Objective

- To study the behavior of a series LCR resonant circuit and to estimate the resonant frequency and Q-factor.
- To study the behavior of voltage drop across inductor and capacitor and hence estimate the resonant frequency.

# 2 Theory and Working Formula

The LCR Circuit consists of a resistor, inductor and a capacitor in series. This circuit when subjected to an AC source can be used to filter-in or filter-out a given frequency of signal. Let us assume the maximum potential across the LCR circuit to be $V_i$ (across pin 1 and 4 with pin 4 as zero potential in figure 1) and the angular frequency of the AC source as $\omega$. The input voltage is thus given by:

$$\mathcal{E}(t) = V_i \sin(\omega t) \tag{1}$$

The impedance of the circuit is given by:

$$|Z| = \sqrt{(R_{DC})^2 + (\mathcal{X}_L - \mathcal{X}_C)^2} \tag{2}$$

where $\mathcal{X}_L$ and $\mathcal{X}_C$ are the inductive and capactive reactance given by:

$$\mathcal{X}_L = \omega L \tag{3}$$

$$\mathcal{X}_C = \frac{1}{\omega C} \tag{4}$$

The output voltage $V_R$ across the resistor (between pin 3 and 4 with pin 4 at zero potential) is given by:

$$V_R = V_i \left(\frac{R}{Z}\right) \sin(\omega t - \phi_R) \tag{5}$$

where $\phi_R$ is the phase difference given by:

$$\phi_R = \arctan\left(\frac{\mathcal{X}_L - \mathcal{X}_C}{R_{DC}}\right) \tag{6}$$

The potential difference across the resistance depends on the impedance of the circuit which depends on the frequency of the AC supply. In the resonance condition, the inductive and capacitative reactance are equal which is when the impedance is minimum. Hence, $V_R$ is maximum in the resonance condition when inductive and capacitative reactance are equal which is true for the resonance frequency $f_o$:

$$f_o = \frac{1}{2\pi\sqrt{LC}} \tag{7}$$

Hence, one can measure the "strength" of a Fourier mode in a given input signal using an LCR circuit by measuring the $V_R$. For thiss reason, LCR circuits are also known as "acceptor circuits" as they allow only a given band of frequencies around the resonance frequency to pass through in large amounts but block the other frequencies to an extend. Similarly, if one wants to filter-out (remove) a given frequency of signal from the input signal, one can take in output voltage across the inductor and the capacitor (excluding the resistor) as:

$$V_{LC} = V_i \left(\frac{\mathcal{X}_L - \mathcal{X}_C}{Z}\right) \sin(\omega t - \phi_{LC}) \tag{8}$$

where $\phi_{LC}$ is given by:

$$\phi_{LC} = -\arctan\left(\frac{R_{DC}}{\mathcal{X}_L - \mathcal{X}_C}\right) \tag{9}$$

Hence, $V_{LC}$ is minimum in the resonance condition wheras it increase as the frequency of the AC supply is far away from the resonance condition.

The ratio of the output power through the resistor to the input voltage ($V_R/V_i$) and the ratio of the output voltage across the inductor and capacitor series to the input voltage ($V_{LC}/V_i$) is given by:

$$\frac{V_R(\text{pp})}{V_i(\text{pp})} = \frac{R}{Z} \tag{10}$$

$$\frac{V_{LC}(\text{pp})}{V_i(\text{pp})} = \frac{\sqrt{(\mathcal{X}_L - \mathcal{X}_C)^2 + R^2_{\text{inductor}}}}{Z} \tag{11}$$

The Q-factor of an LCR is a measure of how good the circuit is in filtering out a given frequency, that is, how narrow is the graph of output voltage with respect to frequency. It is given by:

$$Q = \frac{f_o}{f_2 - f_1} = \frac{1}{\omega_o R_{DC} C} \tag{12}$$

where $f_o$ is the resonant frequency ($\omega_o = 2\pi f_o$) while $f_2$ and $f_1$ are the frequencies where $V_R/V_i$ is equal to $1/\sqrt{2}$ with $f_2 > f_1$.

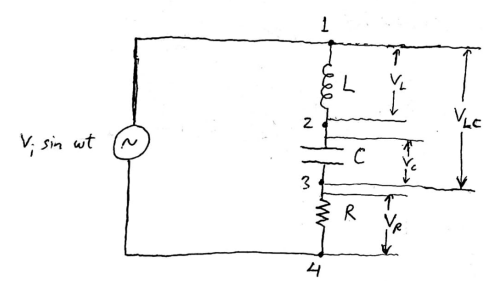



Figure 1: Circuit diagram of the LCR circuit

# 3 Materials Required

(i) Function generator, (ii) Oscilloscope, (iii) Multimeter (iv) Resistor (50Ω and 100Ω), (v) Inductor (10mH), (vi) Capacitor (100 nF), (vii) Breadboard, (viii) Connecting wires.

# 4 Observations

L = 10.408 mH, C = 97.6 nF, $f_o = \frac{1}{2\pi\sqrt{LC}} = 4993.6\,\text{Hz} = 4.994\,\text{kHz}$
Internal resistance of inductor = 29.2Ω
Output impedance of function generator = 50Ω

Table 1: $R_1 = 46.6\,\Omega$

| **Sl. No.** | **Frequency (in kHz)** | $V_i$**(pp) (in V)** | $V_r$**(pp) (in V)** | $\phi_R$ **(in deg)** | $V_r$**(pp)/**$V_i$**(pp)** | $V_r$**(pp)/**$V_i$**(pp) (calculated)** | $\phi_R$ **(in deg) (calculated)** |
|---|---|---|---|---|---|---|---|
| 1 | 1.984 | 4.88 | 0.328 | -80 | 0.06 | 0.07 | -83 |
| 2 | 2.427 | 4.88 | 0.432 | -80 | 0.08 | 0.09 | -81 |
| 3 | 2.604 | 4.80 | 0.480 | -78 | 0.10 | 0.10 | -80 |
| 4 | 2.747 | 4.80 | 0.520 | -79 | 0.10 | 0.11 | -79 |
| 5 | 3.012 | 4.72 | 0.608 | -78 | 0.12 | 0.13 | -77 |
| 6 | 3.205 | 4.64 | 0.680 | -73 | 0.14 | 0.15 | -75 |
| 7 | 3.521 | 4.56 | 0.808 | -70 | 0.17 | 0.19 | -71 |
| 8 | 3.846 | 4.24 | 0.992 | -60 | 0.23 | 0.25 | -66 |
| 9 | 4.032 | 4.00 | 1.12 | -58 | 0.28 | 0.29 | -61 |
| 10 | 4.237 | 3.76 | 1.25 | -55 | 0.33 | 0.35 | -54 |
| 11 | 4.386 | 3.60 | 1.38 | -44 | 0.38 | 0.41 | -48 |
| 12 | 4.464 | 3.36 | 1.40 | -38 | 0.41 | 0.44 | -44 |
| 13 | 4.545 | 3.36 | 1.44 | -32 | 0.42 | 0.48 | -39 |
| 14 | 4.630 | 3.28 | 1.50 | -26 | 0.45 | 0.51 | -33 |
| 15 | 4.717 | 3.28 | 1.52 | -20 | 0.46 | 0.55 | -26 |
| 16 | 4.808 | 3.12 | 1.54 | -13 | 0.49 | 0.58 | -18 |
| 17 | 4.902 | 3.04 | 1.54 | -6 | 0.50 | 0.61 | -9 |
| 18 | 4.950 | 3.04 | 1.56 | 0 | 0.51 | 0.61 | -4 |
| 19 | 5.000 | 3.04 | 1.60 | 3 | 0.52 | 0.61 | 0 |
| 20 | 5.102 | 3.12 | 1.54 | 7 | 0.49 | 0.60 | 10 |
| 21 | 5.208 | 3.20 | 1.50 | 15 | 0.46 | 0.58 | 19 |
| 22 | 5.319 | 3.28 | 1.50 | 22 | 0.45 | 0.54 | 28 |
| 23 | 5.495 | 3.36 | 1.42 | 35 | 0.42 | 0.47 | 39 |
| 24 | 5.682 | 3.60 | 1.34 | 41 | 0.37 | 0.41 | 48 |
| 25 | 5.747 | 3.68 | 1.30 | 45 | 0.35 | 0.39 | 50 |
| 26 | 5.882 | 3.76 | 1.24 | 46 | 0.33 | 0.35 | 54 |
| 27 | 6.024 | 3.92 | 1.18 | 52 | 0.30 | 0.32 | 58 |
| 28 | 6.329 | 4.24 | 1.06 | 59 | 0.25 | 0.27 | 64 |
| 29 | 6.667 | 4.32 | 0.98 | 63 | 0.22 | 0.23 | 68 |
| 30 | 6.757 | 4.40 | 0.94 | 60 | 0.21 | 0.22 | 69 |
| 31 | 7.042 | 4.40 | 0.88 | 70 | 0.20 | 0.19 | 71 |
| 32 | 7.463 | 4.56 | 0.78 | 69 | 0.17 | 0.17 | 74 |
| 33 | 7.692 | 4.56 | 0.74 | 74 | 0.16 | 0.15 | 75 |
| 34 | 7.813 | 4.56 | 0.70 | 73 | 0.15 | 0.15 | 75 |
| 35 | 8.065 | 4.64 | 0.68 | 75 | 0.14 | 0.14 | 76 |
| 36 | 8.333 | 4.72 | 0.66 | 78 | 0.14 | 0.13 | 77 |
| 37 | 8.621 | 4.72 | 0.60 | 80 | 0.12 | 0.12 | 78 |

Table 2: $R_2 = 98.6\,\Omega$

| **Sl. No.** | **Frequency (in kHz)** | $V_i$**(pp) (in V)** | $V_r$**(pp) (in V)** | $V_r$**(pp)/**$V_i$**(pp)** | $V_r$**(pp)/**$V_i$**(pp) (calculated)** |
|---|---|---|---|---|---|
| 1 | 2.000 | 4.88 | 0.680 | 0.13 | 0.14 |
| 2 | 2.404 | 4.88 | 1.00 | 0.20 | 0.18 |
| 3 | 2.500 | 4.88 | 1.04 | 0.21 | 0.19 |
| 4 | 2.604 | 4.88 | 1.08 | 0.22 | 0.20 |
| 5 | 2.717 | 4.88 | 1.16 | 0.23 | 0.22 |
| 6 | 3.049 | 4.72 | 1.32 | 0.28 | 0.27 |
| 7 | 3.333 | 4.64 | 1.48 | 0.31 | 0.32 |
| 8 | 3.571 | 4.56 | 1.72 | 0.37 | 0.38 |
| 9 | 3.676 | 4.48 | 1.76 | 0.39 | 0.41 |
| 10 | 3.846 | 4.24 | 1.96 | 0.46 | 0.45 |
| 11 | 3.968 | 4.24 | 2.00 | 0.47 | 0.49 |
| 12 | 4.098 | 4.08 | 2.12 | 0.52 | 0.54 |
| 13 | 4.310 | 3.92 | 2.24 | 0.57 | 0.61 |
| 14 | 4.464 | 3.92 | 2.32 | 0.59 | 0.66 |
| 15 | 4.545 | 3.84 | 2.36 | 0.61 | 0.69 |
| 16 | 4.673 | 3.76 | 2.40 | 0.63 | 0.73 |
| 17 | 4.808 | 3.68 | 2.44 | 0.66 | 0.75 |
| 18 | 4.902 | 3.60 | 2.44 | 0.67 | 0.76 |
| 19 | 5.000 | 3.60 | 2.44 | 0.67 | 0.77 |
| 20 | 5.102 | 3.68 | 2.44 | 0.66 | 0.76 |
| 21 | 5.208 | 3.68 | 2.44 | 0.66 | 0.75 |
| 22 | 5.319 | 3.76 | 2.40 | 0.63 | 0.73 |
| 23 | 5.435 | 3.84 | 2.36 | 0.61 | 0.70 |
| 24 | 5.682 | 3.92 | 2.28 | 0.58 | 0.64 |
| 25 | 5.814 | 3.92 | 2.20 | 0.56 | 0.60 |
| 26 | 6.098 | 4.08 | 2.08 | 0.51 | 0.53 |
| 27 | 6.250 | 4.24 | 2.04 | 0.48 | 0.50 |
| 28 | 6.579 | 4.24 | 1.88 | 0.44 | 0.44 |
| 29 | 6.944 | 4.48 | 1.72 | 0.38 | 0.38 |
| 30 | 7.246 | 4.56 | 1.60 | 0.35 | 0.35 |
| 31 | 7.463 | 4.56 | 1.56 | 0.34 | 0.33 |
| 32 | 7.692 | 4.64 | 1.44 | 0.31 | 0.31 |
| 33 | 8.475 | 4.80 | 1.28 | 0.26 | 0.25 |
| 34 | 8.772 | 4.80 | 1.24 | 0.25 | 0.24 |
| 35 | 9.091 | 4.88 | 1.16 | 0.23 | 0.22 |
| 36 | 9.615 | 4.88 | 1.08 | 0.22 | 0.20 |
| 37 | 10.20 | 4.88 | 1.04 | 0.21 | 0.18 |

Table 3: $R_2 = 98.6\,\Omega$

| **Sl. No.** | **Frequency (in kHz)** | $V_i$**(pp) (in V)** | $V_{LC}$**(pp) (in V)** | $\phi_{LC}$ **(in deg)** | $\frac{V_{LC}(\mathrm{pp})}{V_i(\mathrm{pp})}$ | $\frac{V_{LC}(\mathrm{pp})}{V_i(\mathrm{pp})}$ **(calculated)** | $\phi_{LC}$ **(in deg) (calculated)** |
|---|---|---|---|---|---|---|---|
| 1 | 1.163 | 4.88 | 4.80 | 5 | 0.98 | 1.00 | 5 |
| 2 | 1.397 | 4.88 | 4.80 | 6 | 0.98 | 0.99 | 6 |
| 3 | 1.572 | 4.88 | 4.80 | 6 | 0.98 | 0.99 | 7 |
| 4 | 1.748 | 4.88 | 4.72 | 8 | 0.97 | 0.99 | 8 |
| 5 | 1.852 | 4.88 | 4.72 | 8 | 0.97 | 0.99 | 9 |
| 6 | 2.016 | 4.88 | 4.64 | 8 | 0.95 | 0.98 | 10 |
| 7 | 2.155 | 4.80 | 4.64 | 9 | 0.97 | 0.98 | 11 |

| Sl. No. | Frequency (in kHz) | $V_i$(pp) (in V) | $V_{LC}$(pp) (in V) | $\phi_{LC}$ (in deg) | $\frac{V_{LC}(\text{pp})}{V_i(\text{pp})}$ | $\frac{V_{LC}(\text{pp})}{V_i(\text{pp})}$ (calculated) | $\phi_{LC}$ (in deg) (calculated) |
|---|---|---|---|---|---|---|---|
| 8 | 2.358 | 4.80 | 4.56 | 10 | 0.95 | 0.97 | 13 |
| 9 | 2.577 | 4.72 | 4.48 | 14 | 0.95 | 0.96 | 15 |
| 10 | 2.874 | 4.72 | 4.40 | 16 | 0.93 | 0.95 | 18 |
| 11 | 2.976 | 4.64 | 4.32 | 17 | 0.93 | 0.94 | 19 |
| 12 | 3.125 | 4.64 | 4.16 | 18 | 0.90 | 0.93 | 21 |
| 13 | 3.788 | 4.24 | 3.36 | 27 | 0.79 | 0.82 | 34 |
| 14 | 4.098 | 4.00 | 2.88 | 29 | 0.72 | 0.71 | 44 |
| 15 | 4.386 | 3.86 | 2.16 | 37 | 0.56 | 0.55 | 56 |
| 16 | 4.545 | 3.68 | 1.92 | 39 | 0.52 | 0.43 | 64 |
| 17 | 4.630 | 3.60 | 1.60 | 37 | 0.44 | 0.36 | 68 |
| 18 | 4.717 | 3.60 | 1.44 | 27 | 0.40 | 0.28 | 73 |
| 19 | 4.808 | 3.60 | 1.28 | 27 | 0.36 | 0.19 | 79 |
| 20 | 4.902 | 3.52 | 1.20 | 21 | 0.34 | 0.09 | 84 |
| 21 | 5.000 | 3.60 | 1.20 | 3 | 0.33 | 0.01 | -89 |
| 22 | 5.051 | 3.60 | 1.24 | -3 | 0.34 | 0.06 | -86 |
| 23 | 5.102 | 3.68 | 1.32 | -14 | 0.36 | 0.11 | -83 |
| 24 | 5.319 | 3.76 | 1.52 | -27 | 0.40 | 0.31 | -72 |
| 25 | 5.495 | 3.84 | 1.88 | -27 | 0.49 | 0.44 | -63 |
| 26 | 5.618 | 3.92 | 2.16 | -32 | 0.55 | 0.52 | -58 |
| 27 | 7.042 | 4.56 | 3.68 | -25 | 0.81 | 0.87 | -29 |
| 28 | 7.634 | 4.56 | 4.00 | -16 | 0.88 | 0.91 | -24 |
| 29 | 8.170 | 4.64 | 4.16 | -16 | 0.90 | 0.93 | -20 |
| 30 | 8.446 | 4.72 | 4.28 | -16 | 0.91 | 0.94 | -19 |
| 31 | 8.929 | 4.88 | 4.36 | -16 | 0.89 | 0.95 | -17 |
| 32 | 9.328 | 4.88 | 4.44 | -13 | 0.91 | 0.96 | -16 |
| 33 | 9.690 | 4.88 | 4.48 | -11 | 0.92 | 0.96 | -15 |
| 34 | 10.08 | 4.88 | 4.56 | -11 | 0.93 | 0.97 | -14 |
| 35 | 10.68 | 4.88 | 4.56 | -12 | 0.93 | 0.97 | -13 |
| 36 | 11.16 | 4.88 | 4.56 | -9 | 0.93 | 0.98 | -12 |

# 5 Results

The theoretical values of $\phi_R$, $\phi_{LC}$, $V_R(\text{pp})/V_i(\text{pp})$ and $V_{LC}(\text{pp})/V_i(\text{pp})$ in tables 1, 2 and 3 were calculated using equations 6, 9, 10 and 11 respectively. The $R_{DC}$ is calculated by adding the resistance of the resistor and the inductor as they are in series.

The plots for the data obtained in the observations is given in figures 2 and 3 along with the theoretical plots.

By fitting (via Scipy, a python module) for the $f$ versus $V_R/V_i$ data in tables 1 and 2 using equation 10, the value for the resonance frequency $f_o$ is obtained as:

$$\text{For R} = 46.6\,\Omega : f_o = 5.143 \pm 0.071\,\text{Hz}$$
$$\text{For R} = 98.6\,\Omega : f_o = 5.143 \pm 0.352\,\text{Hz}$$

Similarly for the Q-factor, fitting was performed using equation 12 as:

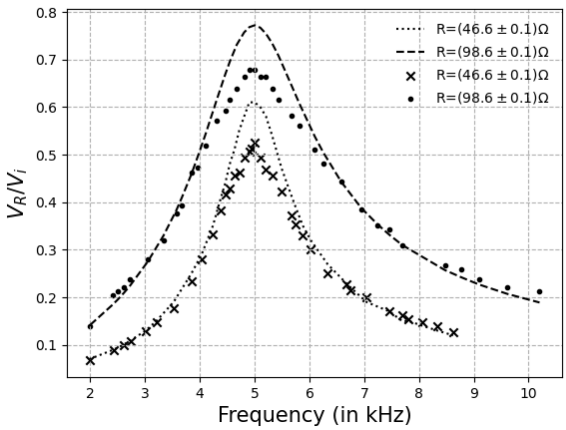
$$\text{For R} = 46.6\,\Omega : Q = 13.277 \pm 1.832$$
$$\text{For R} = 98.6\,\Omega : Q = 13.279 \pm 3.517$$

# 6 Discussion

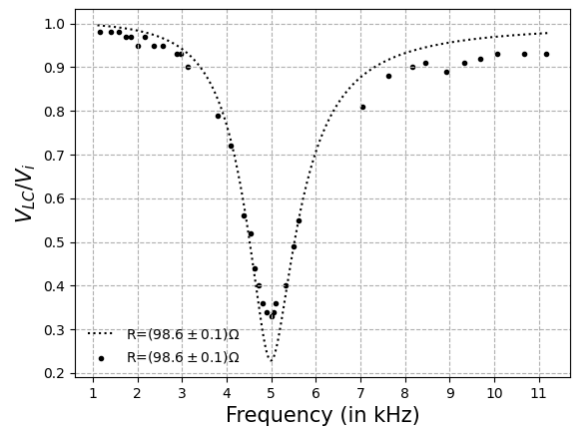
From the plots of $V_R/V_i$ versus frequency, it is clearly seen in figure 2 that the ratio increases with frequency, reaches a maximum around the resonant frequency $f_o$, and then decreases again. This behavior matches the theoretical prediction that at resonance the impedance of the circuit is minimum and the current (hence $V_R$) is maximum. Similarly, the plots of $V_{LC}/V_i$ show a dip around the resonant frequency in figure 2 which is consistent with the theory that the inductor and capacitor voltages cancel each other at resonance.

The experimentally obtained values of the resonant frequency $f_o$ are close to the theoretical value $f_o = 4.994$ kHz calculated from $L$ and $C$. The Q-factor values obtained from curve fitting are consistent with those predicted theoretically. Also, the peak of $V_R/V_i$ is higher for a larger resistance value than a smaller resistance value as seen from figure 2 which is consistent with the theory.

The phase plots show that below resonance, the circuit behaves predominantly capacitive ($\phi_R$ negative), while above resonance, it behaves inductive ($\phi_R$ posi-




(a) $V_R/V_i$ v/s $f$ plot for tables 1 and 2




(b) $V_{LC}/V_i$ v/s $f$ plot for table 3

Figure 2: $V_R/V_i$ v/s $f$ and $V_{LC}/V_i$ v/s $f$ plot for the data in tables 1, 2, 3 with scatter plots for the data and the theoretical plots (in dotted/dashed lines).

tive). At resonance, the phase shift approaches zero, which again agrees with the theoretical expectation. Overall, the experimental data and the theoretical curves are in good agreement.

# 7 Conclusion

The experiment successfully demonstrated the resonance phenomenon in a series LCR circuit. The resonance frequency determined experimentally is close to the theoretical value calculated from the known $L$ and $C$ values. The Q-factor was also obtained both theoretically and experimentally. The measurements of voltage across the resistor and across the LC branch confirmed that $V_R$ attains a maximum and $V_{LC}$ attains a minimum at resonance, which agrees with the theoretical prediction.

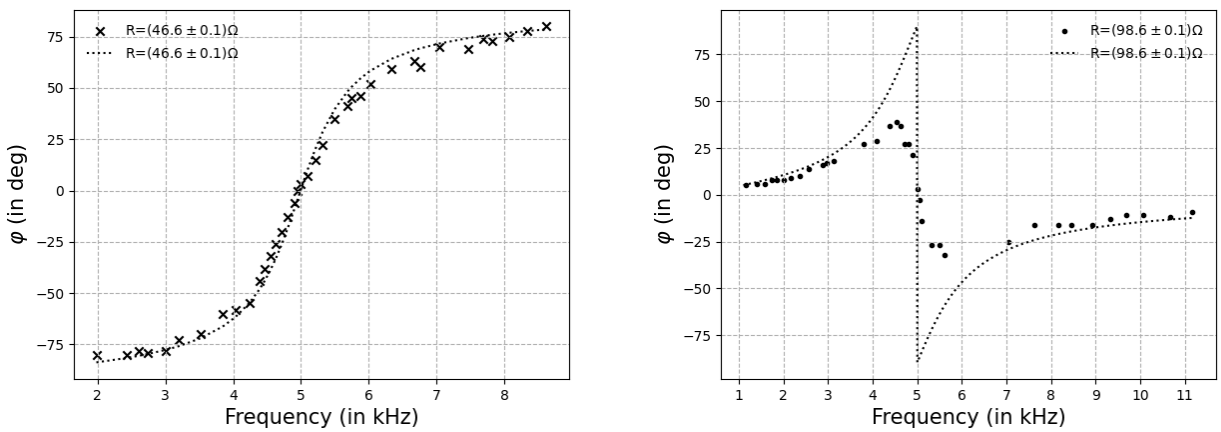
# 8 Precautions

1. The ground of the function generator and the oscilloscope must be connected together to avoid incorrect readings.

2. All connections should be checked carefully to avoid loose contacts which may lead to noisy or fluctuating signals.

3. The frequency should be increased in small steps near the expected resonance to obtain an accurate value of $f_o$ and sharp phase variations.

4. The internal resistance of the inductor must be considered in the calculations.




(a) $\phi_R$ v/s $f$ plot for table 1

(b) $\phi_{LC}$ v/s $f$ plot for table 3

Figure 3: $\phi_R$ v/s $f$ and $\phi_{LC}$ v/s $f$ plot for the data in tables 1 and 3 respectively with scatter plots for the data and the theoretical plots (in dotted/dashed lines).